

NAMIBIA SENIOR SECONDARY CERTIFICATE

BIOLOGY ADVANCED SUBSIDIARY LEVEL

8223/3

PAPER 3 Advanced Practical Skills

2023

CONFIDENTIAL INSTRUCTIONS TO SUBJECT TEACHERS

Great care should be taken to ensure that any confidential information given, including the identity of material on microscope slides where appropriate, does not reach the candidates either directly or indirectly.

This document consists of **5** printed pages and **3** blank pages.



Republic of Namibia

MINISTRY OF EDUCATION, ARTS AND CULTURE

Instruction to teachers

The following fresh materials need to **be obtained in good time** before the practical examination:

- Red/purple onion. Each learner needs a fresh slice of about 3 × 3 cm of red/purple onion – the red/purple epidermis should be intact.
- A small bag (500 g) of table salt.
- Packet of yeast (a 10 g packet is sufficient for ±80 learners).
- A roll of kitchen paper / paper towel – each learner needs 1 piece.

Prepare the day before the examination:

The following will prepare sufficient immobilized enzyme for ±20 learners

1. Prepare a 10% yeast solution by adding 2.5 g of yeast to 25 cm³ of distilled water.
2. Prepare a 2% sodium alginate solution by dissolving 0.5 g sodium alginate in 25 cm³ warm distilled water. Stir continuously until completely dissolved making a thick liquid (this may take quite a while).
3. Prepare a 2% calcium chloride solution by dissolving 1 g calcium chloride granules in 50 cm³ distilled water in a 250 cm³ beaker.
4. In a small beaker add 2 cm³ of yeast solution to 2 cm³ sodium alginate solution and mix well.
5. Draw up the yeast-alginate solution in a 2 cm³ syringe and hold 20 cm above the glass beaker containing the calcium chloride solution.
6. As you carefully press single drops from the syringe into the calcium chloride solution, beads will form. Leave them in the calcium chloride solution for about 5 minutes, after which you remove them with a tea/fine strainer and gently wash them under cold tap water for one minute.

Important: Replace the calcium chloride solution after roughly every 100 beads.

7. Finally rinse them with distilled water and store them in a beaker / ice cream box **with distilled water in the refrigerator** until the day of the practical examination.

To prepare 10% hydrogen peroxide solution from 100 volume hydrogen peroxide, place 50 cm³ hydrogen peroxide into a 250 cm³ beaker and add 100 cm³ of distilled water. Keep in a closed container. This will be sufficient hydrogen peroxide for ± 15 learners.

Prepare a 10% salt solution by dissolving 50 g of salt in 500 cm³ of tap water. This solution will be enough for ± 15 learners. Cut the red/purple onion into pieces of 3 × 3 cm and place them into the salt solution. For further instructions see “Question 2” part of this instruction.

Question 1:

Each candidates requires the following equipment and chemicals:

- 1 × pair of forceps or tweezers
- 1 × stopwatch
- 1 × six-hole test tube rack with 5 test tubes labelled P, Q, R, S and T.
- 5 × 50 cm³ glass beakers labelled P, Q, R, S and T. Beaker P must be filled with 10 cm³ of 10% hydrogen peroxide solution
- 1 × 100 cm³ beaker labelled W and filled with 50 cm³ distilled water
- 5 × 5 cm³ syringes
- 1 × Petri dish labelled “beads” containing 20 yeast-alginate beads

Question 2:

Each candidates requires the following equipment and chemical:

- 1 × light microscope with high-power and low-power objective lenses
- 1 × white tile
- 1 x scalpel
- 1 × 50 cm³ beaker containing a 3 × 3 cm piece of red/purple onion in 10% salt solution labelled “**N1**”
- 1 × microscope slide
- 1 × cover slip
- 1 × plastic dropper pipette
- 1 × piece of kitchen paper / paper towel
- 1 × pair of forceps or tweezers

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REPORT FORM
Republic of Namibia
NAMIBIA SENIOR SECONDARY CERTIFICATE

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Question Paper & Examination Security Declaration

Region.....Date.....

Centre Number.....Centre Name

Principal (full name, print).....

Head of Examination Centre (full name, print).....

Hereby it is certified that this examination was conducted under appropriate measures and conditions as stipulated in the 'Rules & Regulations for the NSSC Examinations'.

Attached are the following details;

1 Subjects Teacher's Results

*Please submit details of the readings obtained in **Question 1** and **Question 2** on a spare copy of the question paper clearly marked '**Subject Teacher's Results**' and showing the Centre Number.*

*It is essential that each packet of candidates' scripts contains a copy of the applicable '**Subject Teacher's Results**' as the candidates' work cannot be assessed accurately without such information.*

2 The Candidate Numbers of candidates attending each session were:

Please write them in their sitting order.

First Session	Second Session	Third Session

- any general difficulties encountered in making preparation
- difficulties due to faulty apparatus or materials
- accident to apparatus or materials
- assistance with respect to colour-blindness

Other cases of hardship, e.g. illness, temporary disability, should be reported directly to DNEA through the regional offices on the normal 'Application for Special Consideration' form.

Report on any difficulties experienced by candidates.

[illegible]

Principal's Signature & Date.....

Head of Examination Centre's Signature & Date:.....

Subject Teacher's Full Name:

Subject Teacher's Signature & Date:

Invigilator/Supervisor's Signature & Date:

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